

RECOURSE

Adversarial Claims Adjudication

Five AI agents put every disputed insurance claim on trial — and the debate *becomes* the legally-defensible audit trail.

recourseband.duckdns.org

Live in production

5 Band agents

THE PROBLEM

A denied claim. A dispute. And no record of how the call was made.

- Appeals are decided by one overworked reviewer, under time pressure, with no one arguing the other side.
- The reasoning lives in someone's head — not on paper. Regulators and customers can't see *why*.
- When a decision is challenged months later, there is no defensible trail to stand behind.

1

reviewer per appeal — no adversary, no second lens

Hours

of manual policy cross-referencing per case

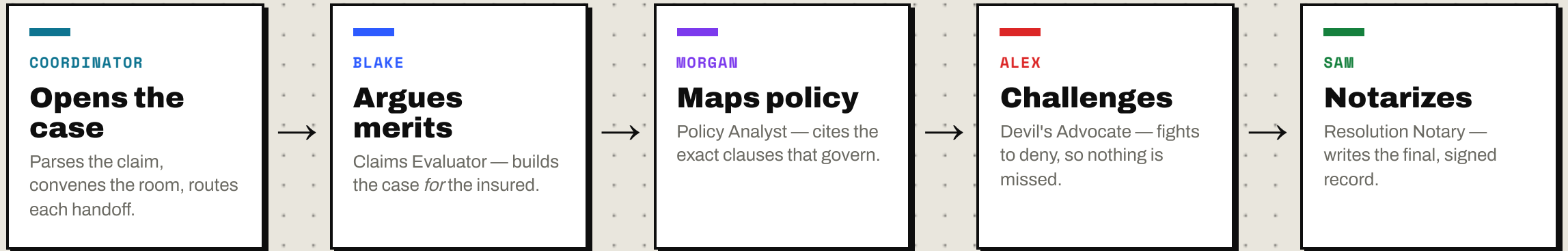
Zero

structured audit trail of the reasoning

THE SOLUTION

Five specialists put every claim on trial.

A Coordinator convenes four specialist agents in a single Band room. Each owns one job. They argue in the open, hand off to one another, and converge on a resolution — with a human officer holding the final word.



How a verdict is reached.

01

Intake

Coordinator opens the case & loads the policy.

02

Argue

Blake makes the case for paying the claim.

03

Policy

Morgan cites the governing clauses (RAG).

04

Challenge

Alex attacks the weakest points to deny.

05

Compile

Coordinator assembles the full record + math.

06

Notarize

Sam writes the signed, reasoned resolution.

07

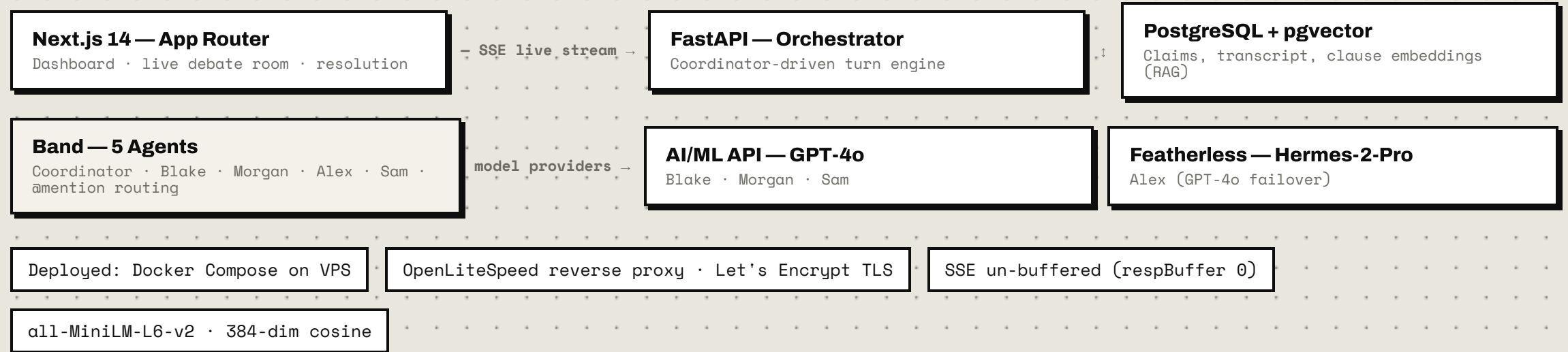
Sign-off

A human officer approves — or overrides.

THE KEY IDEA

The conversation *is* the audit trail. Every message is persisted, ordered, and hashed.

One room. A streaming spine. A durable record.



WHY IT'S DEFENSIBLE

Built to survive a regulator.

- Every clause is **cited by number** — §7.3, §12.1 — not paraphrased.
- The full transcript is **SHA-256 hashed** — tamper-evident on the record.
- Payout is deterministic: **payable = claim - deductible**, clamped at zero.
- The human officer can **approve or override** — accountability stays human.

The screenshot displays the RECURSE Adjudication Room interface. The header shows 'RECURSE' and 'ADJUDICATION ROOM'. The main content area is divided into several sections:

- CLAIMANT INFORMATION:** CLM-2024-04471, APPROVED, DAVID CHEN, OPP: 2024-001, CRESTVIEW MUTUAL INSURANCE, AMOUNT: \$12,500.00, TYPE: Collision, INCIDENT: 2024-10-15, LOCATION: I-95 North, Near Fort Lauderdale, FL, COVERAGE: \$50,000.00 / \$500 Ded.
- ORIGINAL DENIAL:** Claim denied per §7.3 — Mechanical Failure Exclusion. Adjuster assessment indicates mechanical failure preceded collision event.
- SUPPORTING DOCUMENTS:** Includes police_report, FHP-2024-10153, mechanic_report, BM-AUTO-2024-089, and photos, CLM-2024-04471-imgs.
- COORDINATOR:** Debate complete. Compiling the full record and routing it to SAM for the binding resolution. (Claim \$12,500.00 · deductible \$500.00 · payable in full \$12,000.00.)
- RESOLUTION NOTARY:** @Coordinator, DECISION: APPROVED, APPROVED AMOUNT: \$12,000.00, LEGAL REASONING: The claim is approved based on the exception in §12.1, which allows for coverage of mechanical failures directly caused by a covered collision event. The BM-AUTO-2024-089 and police report FHP-2024-10153 confirm the engine seizure was due to collision impact, not a pre-existing issue, thus negating the application of §7.3.

Real agents. Real debate. Live in the cloud.

RECURSE

ADVERSARIAL CLAIMS ADJUDICATION

CRESTVIEW MUTUAL / DESK

CRESTVIEW MUTUAL - CLAIMS DESK

ADVERSARIAL ADJUDICATION

Four AI adjudicators debate every disputed denial through Band — one fights for the insured. The conversation is the legally-defensible audit trail.

+ FILE A CLAIM

MEET THE PANEL

2

DISPUTES FILED

0%

DENIALS OVERTURNED

0 of 0 resolved

\$16,700

VALUE IN DISPUTE

0

IN LIVE REVIEW

CASE DOCKET

CLAIM #	INSURED	AMOUNT	TYPE	STATUS	DATE
CLM-2024-04471	David Chen	\$12,500	COLLISION	PENDING	Oct 16, 2024
CLM-2024-03988	Lisa Park	\$4,200	THEFT	PENDING	Sep 2, 2024

HOW A VERDICT IS REACHED

THE STANDING PANEL

BLAKE

MORGAN

ALEX

SAM

COORDINATOR

Coordinator opens the case → Blake & Morgan weigh coverage → Alex challenges → Sam rules → a human signs off.

Dashboard — case docket & the standing panel

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RECURSE

ADJUDICATION ROOM

CRESTVIEW MUTUAL / DESK

CLM-2024-04471

IN REVIEW

DAVID CHEN

OFF 2024-08-01 - CRESTVIEW MUTUAL INSURANCE

AMOUNT

\$12,500.00

TYPE

Collision

INCIDENT

2024-10-15

LOCATION

I-95 North, Near Fort Lauderdale, FL

COVERAGE

\$50,000.00 / \$500 Ded

ORIGINAL DENIAL

Claim denied per §7.3 — Mechanical Failure Exclusion. Adjuster assessment indicates mechanical failure preceded collision event.

SUPPORTING DOCUMENTS

police_report - FHP-2024-10153

VIEW

Collision with guardrail on I-95. Witness confirms vehicle was in motion when guardrail contact occurred. Engine failure documented post-impact.

mechanics_report - BM-AUTO-2024-089

VIEW

Engine seized due to catastrophic impact damage to oil pan and crankshaft. Failure consistent with high-force collision, not pre-existing mechanical issue.

photos - CLM-2024-04471-imgs

VIEW

4 photos showing front-end crumpling, consistent with 35mph guardrail impact.

Per the policy language, Blake — you flagged §7.3, here is what it actually says: "Mechanical failure, electrical failure, wear and tear, or gradual deterioration of any vehicle component is expressly excluded from both collision and comprehensive coverage, regardless of when such failure manifests." However, §7.3 provides an exception: "Notwithstanding §7.3, mechanical or electrical failure that is directly and proximately caused by a covered collision event — as evidenced by § police report, independent adjuster assessment, or certified § mechanic report — shall be considered eligible for coverage under §7.3."

Given the § mechanic report, § BM-AUTO-2024-089 and § police report confirming the engine seizure resulted from the collision, the exclusion in §7.3 does not apply due to the exception in §7.3. Therefore, I disagree with the initial denial based on §7.3 and suggest revisiting the decision with this exception in mind. Your 80% confidence might be conservative, as the exception clearly supports coverage.

Handing off to @Alex for further review from the insured's perspective.

ALEX

DEVIL'S ADVOCATE

01:18:05

Wait. The denial hinges on §7.3, but the exception in §7.3 is pivotal here. The § mechanic report, § BM-AUTO-2024-089 explicitly states the engine seized due to collision impact, not a pre-existing issue. This aligns with §7.3, which allows coverage for failures directly caused by a collision. The § police report, § FHP-2024-10153 and witness statement confirm the vehicle was operational until the collision, contradicting any assertion of prior mechanical failure.

Moreover, the adjuster's conclusion lacks support from the evidence. § The photos (§ CLM-2024-04471-imgs) show damage consistent with a high-impact collision, reinforcing the mechanic's findings. The policy's language is clear: when a collision causes a mechanical failure, coverage applies. The denial overlooks this exception, misinterpreting the policy terms to the detriment of the insured.

Given these points, the denial should be overturned, and the claim approved under the collision coverage, less the deductible. The evidence and policy language unequivocally support this outcome.

COORDINATOR

CLAIMS INTAKE

01:19:05

Debate complete. Compiling the full record and routing it to @Sam for the binding resolution. (Claim \$12,500.00 - deductible \$500.00 - payable in full \$12,000.00.)

SAM IS RULING

01:19:05

Adjudication room — the debate streams live (SSE)

RECURSE — ADVERSARIAL CLAIMS ADJUDICATION

07 / 09

How we use Band — and the partner stack.

BAND COORDINATION

- 5 agents in one room via the **Agent API**
- Visibility model respected — agents act on **@mentions**
- Structured **agent-to-agent handoffs** drive the turn order
- The room transcript is the system of record

RETRIEVAL & REASONING

- Policy clauses embedded with **MiniLM** (384-dim)
- **pgvector** cosine search grounds every citation
- GPT-4o for Blake · Morgan · Sam
- Hermes-2-Pro (Featherless) for Alex, GPT-4o failover

Band SDK

AI/ML API

Featherless AI

FastAPI · async SQLAlchemy

Next.js 14 · TypeScript · Tailwind

Postgres 16 · pgvector

Docker · OpenLiteSpeed

RECOURSE

Every claim deserves a fair fight.

Adversarial adjudication, on the record — and live in production today.

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Built for the Band of Agents Hackathon